

**This Practicals is
performed and
created by :**

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Practical-1

AIM: Install Android Studio, set up the Android development environment, and create a simple "Hello World" app.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:layout_gravity="center"/>
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

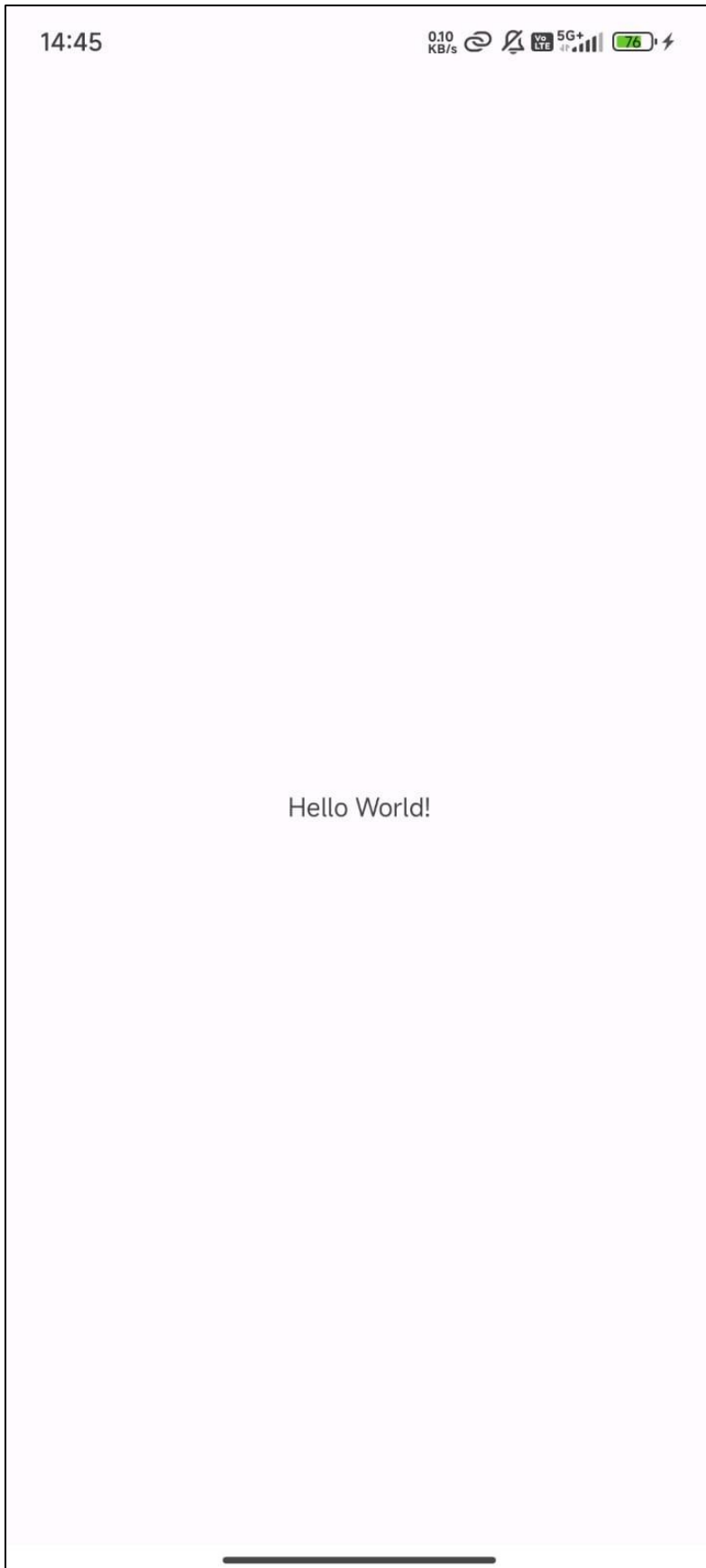
import android.os.Bundle;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical1 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical1);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        Toast.makeText(this, "welcome to Practical 1", Toast.LENGTH_SHORT).show();
    }
}
```

OUTPUT:



Practical-2

AIM: Develop a simple app that demonstrates the activity lifecycle.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:layout_gravity="center"/>
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.os.Handler;
import android.util.Log;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical2);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        Log.d("PRACTICAL2", "On Create called");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("PRACTICAL2", "On Start called");
    }
}
```

```

}

@Override
protected void onResume() {
    super.onResume();
    Log.d("PRACTICAL2", "On Resume called");
}

@Override
protected void onPause() {
    super.onPause();
    Log.d("PRACTICAL2", "On Pause called");
}

@Override
protected void onStop() {
    super.onStop();
    Log.d("PRACTICAL2", "On Stop called");
}

@Override
protected void onRestart() {
    super.onRestart();
    Log.d("PRACTICAL2", "On Restart called");
}

@Override
protected void onDestroy() {
    super.onDestroy();
    Log.d("PRACTICAL2", "On destroy called");
}
}

```

OUTPUT

2024-09-12 14:41:47.344	23833-23833	PRACTICAL2	com.mk.madpractical	D	On Create called
2024-09-12 14:41:47.350	23833-23833	PRACTICAL2	com.mk.madpractical	D	On Start called
2024-09-12 14:41:47.353	23833-23833	PRACTICAL2	com.mk.madpractical	D	On Resume called
2024-09-12 14:41:47.406	23833-23833	VRI[Practical2]	com.mk.madpractical	D	vri.reportNextDraw android.view.ViewRootImpl.performTraversals:4293 android.view.ViewRoo
2024-09-12 14:41:47.406	23833-23833	VRI[Practical2]	com.mk.madpractical	D	vri.Setup new sync=wmsSync-VRI[Practical2]#2
2024-09-12 14:41:47.414	23833-23833	VRI[Practical2]	com.mk.madpractical	D	vri.reportDrawFinished
2024-09-12 14:41:54.745	23833-23833	MIUIInput	com.mk.madpractical	D	[KeyEvent] ViewRootImpl windowName 'sendBackKeyEvent-com.mk.madpractical/com.mk.madpract
2024-09-12 14:41:54.749	23833-23833	MIUIInput	com.mk.madpractical	D	[KeyEvent] ViewRootImpl windowName 'sendBackKeyEvent-com.mk.madpractical/com.mk.madpract
2024-09-12 14:41:54.774	23833-23833	PRACTICAL2	com.mk.madpractical	D	On Pause called
2024-09-12 14:41:55.417	23833-23833	PRACTICAL2	com.mk.madpractical	D	On Stop called
2024-09-12 14:41:55.421	23833-23833	PRACTICAL2	com.mk.madpractical	D	On destroy called

Logcat Output

Practical-3

AIM: Develop a simple calculator app that takes user input and performs basic arithmetic operations like addition, subtraction, multiplication, and division.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp"
    android:gravity="center">

    <EditText
        android:id="@+id/result"
        android:inputType="text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Result"
    />

    <EditText
        android:id="@+id/edt_1"
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
    />

    <EditText
        android:id="@+id/edt_2"
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
    />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center">

        <Button
            android:id="@+id/btn_plus"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="+"
            android:textSize="30sp"/>
```

```
<Button
    android:id="@+id/btn_minus"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="-"
    android:textSize="30sp"/>
```

```
<Button
    android:id="@+id/btn_multi"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="x"
    android:textSize="30sp"/>
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center">
    <Button
        android:id="@+id/btn_divi"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="/"
        android:textSize="30sp"/>
    <Button
        android:id="@+id/btn_modu"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="%"
        android:textSize="30sp"/>
```

```
</LinearLayout>
</LinearLayout>
```

JAVA:

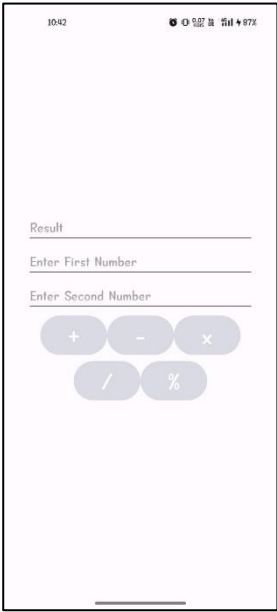
```
package com.mk.madpractical;
import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

```

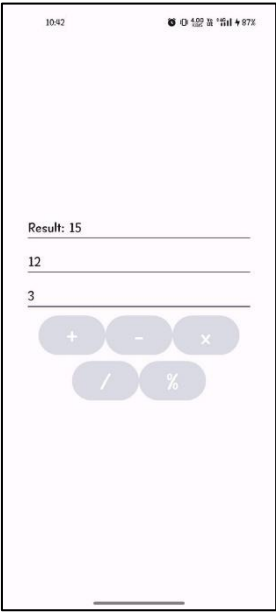
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class Practical3 extends AppCompatActivity {
    EditText n1, n2, Result;
    Button plus, minus, multi, division, modulus;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical3);
        n1 = (EditText) findViewById(R.id.edt_1);
        n2 = (EditText) findViewById(R.id.edt_2);
        plus = (Button) findViewById(R.id.btn_plus);
        minus = (Button) findViewById(R.id.btn_minus);
        multi = (Button) findViewById(R.id.btn_multi);
        division = (Button) findViewById(R.id.btn_divi);
        modulus = (Button) findViewById(R.id.btn_modu);
        Result = (EditText) findViewById(R.id.result);
        View.OnClickListener listener = new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int num1 = 0, num2 = 0, ans = 0;
                try {
                    num1 = Integer.parseInt(n1.getText().toString());
                    num2 = Integer.parseInt(n2.getText().toString());
                } catch (NumberFormatException e) {
                    Toast.makeText(Practical3.this, "Please enter a valid number for both", Toast.LENGTH_SHORT).show();
                }
                if (v.equals(plus)) {
                    ans = num1 + num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(minus)) {
                    ans = num1 - num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(multi)) {
                    ans = num1 * num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(division)) {
                    ans = num1 / num2;
                    Result.setText("Result: " + ans);
                } else if (v.equals(modulus)) {
                    ans = num1 % num2;
                    Result.setText("Result: " + ans);
                }
            }
        };
        plus.setOnClickListener(listener);
        minus.setOnClickListener(listener);
        multi.setOnClickListener(listener);
        division.setOnClickListener(listener);
        modulus.setOnClickListener(listener);
    }
}

```

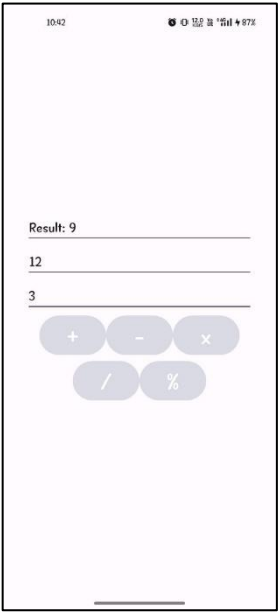

OUTPUT:



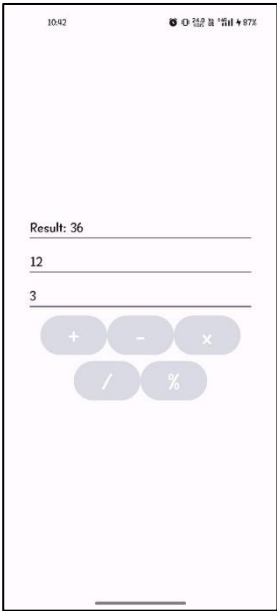
Activity layout



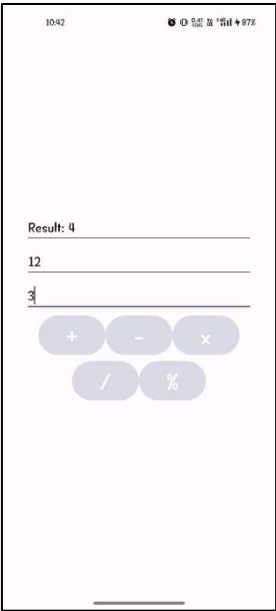
Plus operation



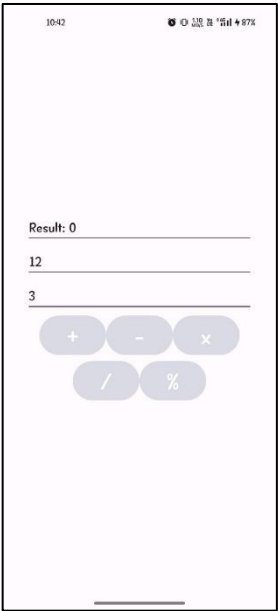
minus operation



Multiplication operation



Division operation



Modulus operation

Practical-4

AIM: Develop an Android application that uses LinearLayout to arrange UI components vertically or horizontally

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

    android:id="@+id/Practical_4_LinearLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 1"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 2"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="vertical align 3"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 1"
            android:layout_margin="10dp"/>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 2"
            android:layout_margin="10dp"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="horizontal align 3"
            android:layout_margin="10dp"/>

    </LinearLayout>

</LinearLayout>

</LinearLayout>
```

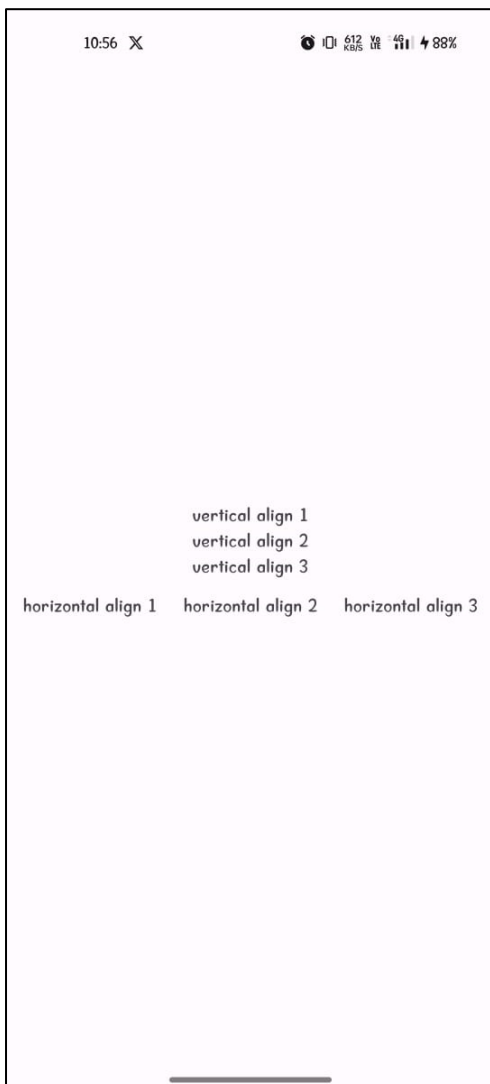
```

JAVA: package com.mk.madpractical;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical4 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical4);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical_4_LinearLayout), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}

```

OUTPUT



Practical-5

AIM: Develop an Android application that uses RelativeLayout to arrange UI components relative to each other.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_width="match_parent"
    android:id="@+id/Practical_5_RelativeLayout"
    android:layout_height="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <TextView
        android:id="@+id/centerText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Center text view" />
    <TextView
        android:id="@+id/topText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_above="@+id/centerText"
        android:layout_marginBottom="25dp"
        android:layout_centerHorizontal="true"
        android:text="Top of center text view" />
    <TextView
        android:id="@+id/Bottom_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/centerText"
        android:layout_marginTop="25dp"
        android:layout_centerHorizontal="true"
        android:text="Bottom of center text view" />
    <TextView
        android:id="@+id/Left_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="30dp"
        android:layout_centerVertical="true"
        android:layout_toStartOf="@+id/centerText"
        android:text="Left of text view" />
    <TextView
        android:id="@+id/Right_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="30dp"
        android:layout_centerVertical="true"
        android:layout_toEndOf="@+id/centerText"
        android:text="Right of text view" />

</RelativeLayout>
```

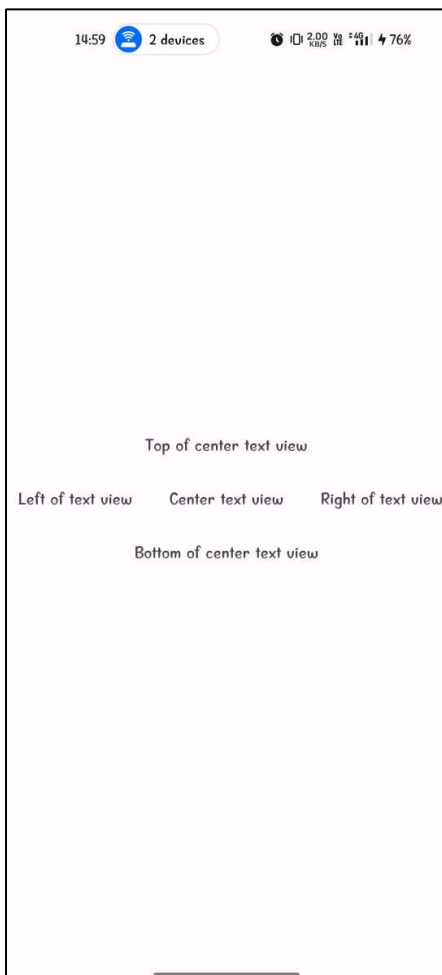
JAVA:

```
package com.mk.madpractical;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical5 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical5);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical_5_RelativeLayout), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

OUTPUT



Custom view for practical 6

XML: `<?xml version="1.0" encoding="utf-8"?>`

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/My_Custom_View_Layout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">
    <ImageView
        android:id="@+id/customImage"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_gravity="center_vertical"
        android:src="@drawable/ic_launcher_foreground"
        android:scaleType="centerCrop" />
    <TextView
        android:id="@+id/customText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_gravity="center_vertical"
        android:text="Custom test here"
        android:layout_marginStart="10dp" />
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Rect;
import android.util.AttributeSet;
import android.view.LayoutInflater;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.TextView;

import java.util.Random;

public class MyCustomView extends LinearLayout {

    private TextView customText;
    private ImageView customImage;

    public MyCustomView(Context context) {
        super(context);
        init(context);
    }
}
```

```

public MyCustomView(Context context, AttributeSet attrs) {
    super(context, attrs);
    init(context);
}

public MyCustomView(Context context, AttributeSet attrs, int defStyleAttr) {
    super(context, attrs, defStyleAttr);
    init(context);
}

private void init(Context context) {
    LayoutInflater inflater = LayoutInflater.from(context);
    inflater.inflate(R.layout.custom_layout, this, true);
    customText = findViewById(R.id.customText);
    customImage = findViewById(R.id.customImage);
}

public void setCustomText(String text) {
    customText.setText(text);
}

public void setCustomImage(int resourceId) {
    customImage.setImageResource(resourceId);
}

public void setCustomImage (Bitmap bitmap){
    customImage.setImageBitmap(bitmap);
}

public Bitmap createBitmapWithLetter(String letter, int color) {
    if (letter.length() > 3){
        letter = letter.substring(0,3);
    }
    Bitmap bitmap = Bitmap.createBitmap(100, 100, Bitmap.Config.ARGB_8888);
    Canvas canvas = new Canvas(bitmap);
    canvas.drawColor(color);
    Paint paint = new Paint();
    paint.setColor(Color.WHITE);
    paint.setTextSize(50);
    paint.setTextAlign(Paint.Align.CENTER);

    Rect bounds = new Rect();
    paint.getTextBounds(letter, 0, letter.length(), bounds);
    int x = (bitmap.getWidth() / 2);
    int y = (int) ((bitmap.getHeight() / 2) - ((paint.descent() + paint.ascent()) / 2));

    canvas.drawText(letter, x, y, paint);

    return bitmap;
}

public int getRandomColor() {
    Random random = new Random();
    return Color.rgb(random.nextInt(256), random.nextInt(256), random.nextInt(256));
}
}

```

Practical-6

AIM: Develop an Android application that uses ScrollView to display a long list of items.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical6">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:id="@+id/Pr6_ScrollView_LinearLayout"
            android:orientation="vertical" />
    </ScrollView>
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.view.Gravity;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.LinearLayout;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical6 extends AppCompatActivity {
    LinearLayout layout;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```



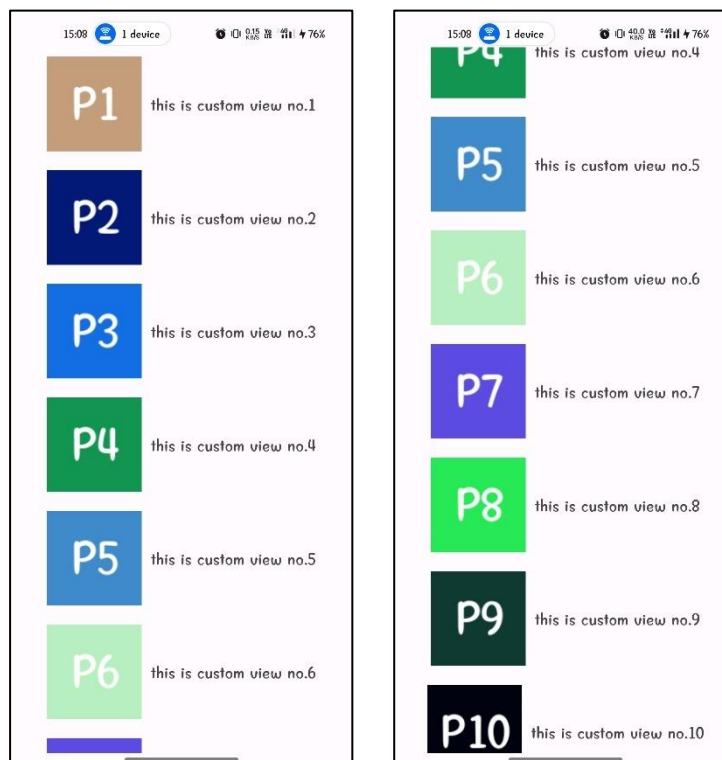
```

EdgeToEdge.enable(this);
setContentView(R.layout.activity_practical6);
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});
layout = findViewById(R.id.Pr6_ScrollView_LinearLayout);
for (int i = 1; i < 31; i++) {
    MyCustomView view = new MyCustomView(this);
    view.setCustomImage(view.createBitmapWithLetter("P"+i,view.getRandomColor()));
    view.setCustomText("this is custom view no."+i);
    LinearLayout lay = (LinearLayout) view.findViewById(R.id.My_Custom_View_Layout);
    lay.setGravity(Gravity.CENTER);
    ImageView imageView = view.findViewById(R.id.customImage);
    int newWidthInDp = 100;
    int newHeightInDp = 100;

    float scale = getResources().getDisplayMetrics().density;
    int newWidthInPx = (int) (newWidthInDp * scale + 0.5f);
    int newHeightInPx = (int) (newHeightInDp * scale + 0.5f);

    ViewGroup.LayoutParams params = imageView.getLayoutParams();
    params.width = newWidthInPx;
    params.height = newHeightInPx;
    imageView.setLayoutParams(params);
    layout.addView(view);
}
}
}

```



Custom view for practical 7

XML: `<?xml version="1.0" encoding="utf-8"?>`

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/My_Custom_View_Layout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">
    <ImageView
        android:id="@+id/customImage"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_gravity="center_vertical"
        android:src="@drawable/ic_launcher_foreground"
        android:scaleType="centerCrop" />
    <TextView
        android:id="@+id/customText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_gravity="center_vertical"
        android:text="Custom test here"
        android:layout_marginStart="10dp" />
</LinearLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.graphics.Rect;
import android.util.AttributeSet;
import android.view.LayoutInflater;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.TextView;

import java.util.Random;

public class MyCustomView extends LinearLayout {

    private TextView customText;
    private ImageView customImage;

    public MyCustomView(Context context) {
        super(context);
        init(context);
    }
}
```

```

public MyCustomView(Context context, AttributeSet attrs) {
    super(context, attrs);
    init(context);
}

public MyCustomView(Context context, AttributeSet attrs, int defStyleAttr) {
    super(context, attrs, defStyleAttr);
    init(context);
}

private void init(Context context) {
    LayoutInflater inflater = LayoutInflater.from(context);
    inflater.inflate(R.layout.custom_layout, this, true);
    customText = findViewById(R.id.customText);
    customImage = findViewById(R.id.customImage);
}

public void setCustomText(String text) {
    customText.setText(text);
}

public void setCustomImage(int resourceId) {
    customImage.setImageResource(resourceId);
}

public void setCustomImage (Bitmap bitmap){
    customImage.setImageBitmap(bitmap);
}

public Bitmap createBitmapWithLetter(String letter, int color) {
    if (letter.length() > 3){
        letter = letter.substring(0,3);
    }
    Bitmap bitmap = Bitmap.createBitmap(100, 100, Bitmap.Config.ARGB_8888);
    Canvas canvas = new Canvas(bitmap);
    canvas.drawColor(color);
    Paint paint = new Paint();
    paint.setColor(Color.WHITE);
    paint.setTextSize(50);
    paint.setTextAlign(Paint.Align.CENTER);

    Rect bounds = new Rect();
    paint.getTextBounds(letter, 0, letter.length(), bounds);
    int x = (bitmap.getWidth() / 2);
    int y = (int) ((bitmap.getHeight() / 2) - ((paint.descent() + paint.ascent()) / 2));

    canvas.drawText(letter, x, y, paint);

    return bitmap;
}

public int getRandomColor() {
    Random random = new Random();
    return Color.rgb(random.nextInt(256), random.nextInt(256), random.nextInt(256));
}
}

```

Practical-7

AIM: Develop an Android application that uses ListView and Custom Adapter to display a list of images with text.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical7">
    <ListView
        android:id="@+id/PR7_ListView"
        android:layout_width="300dp"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;
import android.content.Context; import android.graphics.*; import android.os.Bundle;
import android.view.LayoutInflater; import android.view.View; import android.view.ViewGroup;
import android.widget.BaseAdapter; import android.widget.ImageView; import android.widget.ListView;
import android.widget.TextView; import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import java.util.Random;

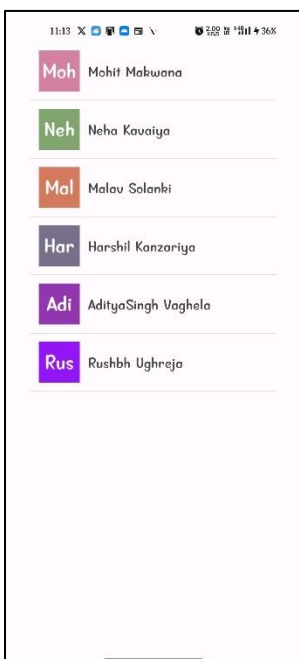
public class Practical7 extends AppCompatActivity {
    ListView listView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical7);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        listView = findViewById(R.id.PR7_ListView);
        String[] arr = {"Mohit Makwana", "Neha Kavaia", "Malav Solanki", "Harshil Kanzariya", "AdityaSingh Vaghela",
        "Rushbh Ughreja"};
        listView.setAdapter(new Pr7CustomAdapter(this, arr));
    }
}
```

```

class Pr7CustomAdapter extends BaseAdapter {
    private Context context;
    private String[] items;
    public Pr7CustomAdapter(Context context, String[] items) {
        this.context = context;
        this.items = items;
    }
    @Override
    public int getCount() {
        return items.length;
    }
    @Override
    public Object getItem(int position) {
        return items[position];
    }
    @Override
    public long getItemId(int position) {
        return position;
    }
    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        MyCustomView view;
        if (convertView == null) {
            view = new MyCustomView(context);
        } else {
            view = (MyCustomView) convertView;
        }
        String item = (String) getItem(position);
        view.setCustomText(item);
        view.setCustomImage(view.createBitmapWithLetter(item, view.getRandomColor()));
        return (View) view;
    }
}

```

OUTPUT



Practical-8

AIM: Develop an Android application that uses the Navigation Drawer to display a side menu.

XML:

Practical_layout file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:openDrawer="start">

    <include
        android:id="@+id/app_bar_practical8"
        layout="@layout/app_bar_practical8"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        android:fitsSystemWindows="true"
        app:headerLayout="@layout/nav_header_practical8"
        app:menu="@menu/activity_main_drawer" />
</androidx.drawerlayout.widget.DrawerLayout>
```

App_bar_layout:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:openDrawer="start">

    <include
        android:id="@+id/app_bar_practical8"
        layout="@layout/app_bar_practical8"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
```

```
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        android:fitsSystemWindows="true"
        app:headerLayout="@layout/nav_header_practical8"
        app:menu="@menu/activity_main_drawer" />
</androidx.drawerlayout.widget.DrawerLayout>
```

content_layout:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

    app:layout_behavior="@string/appbar_scrolling_view_behavi
or"
    tools:showIn="@layout/app_bar_practical8">

    <fragment

        android:id="@+id/nav_host_fragment_content_practical8"

        android:name="androidx.navigation.fragment.NavHostFragme
nt"

        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:defaultNavHost="true"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:navGraph="@navigation/mobile_navigation" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Home_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.home.HomeFragment">

    <TextView
        android:id="@+id/text_home"
        android:layout_width="match_parent"
```

```

    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:layout_marginEnd="8dp"
    android:textAlignment="center"
    android:textSize="20sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Gallery_fragment:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.gallery.GalleryFragment">

    <TextView
        android:id="@+id/text_gallery"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

SlidShow_fragment:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=
"http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.slideshow.SlideshowFragment">

    <TextView
        android:id="@+id/text_slideshow"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"

```

```

        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Activity_main_drawer:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    tools:showIn="navigation_view">

    <group android:checkableBehavior="single">
        <item
            android:id="@+id/nav_home"
            android:icon="@drawable/ic_menu_camera"
            android:title="@string/menu_home" />
        <item
            android:id="@+id/nav_gallery"
            android:icon="@drawable/ic_menu_gallery"
            android:title="@string/menu_gallery" />
        <item
            android:id="@+id/nav_slideshow"
            android:icon="@drawable/ic_menu_slideshow"
            android:title="@string/menu_slideshow" />
    </group>
</menu>

```

Movile_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<navigation
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/mobile_navigation"
    app:startDestination="@+id/nav_home">
    <fragment
        android:id="@+id/nav_home"

        android:name="com.mk.madpractical.ui.home.HomeFragmen
t"
        android:label="@string/menu_home"
        tools:layout="@layout/fragment_home" />
    <fragment
        android:id="@+id/nav_gallery"

        android:name="com.mk.madpractical.ui.gallery.GalleryFragmen
t"
        android:label="@string/menu_gallery"
        tools:layout="@layout/fragment_gallery" />
    <fragment

```

```

        android:id="@+id/nav_slideshow"

        android:name="com.mk.madpractical.ui.slideshow.SlideshowF
        ragment"
        android:label="@string/menu_slideshow"
        tools:layout="@layout/fragment_slideshow" />
    </navigation>

```

Nav_header_layout:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="@dimen/nav_header_height"
    android:background="@drawable/side_nav_bar"
    android:gravity="bottom"
    android:orientation="vertical"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:theme="@style/ThemeOverlay.AppCompat.Dark">
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:contentDescription="@string/nav_header_desc"

        android:paddingTop="@dimen/nav_header_vertical_spacing"
        app:srcCompat="@mipmap/ic_launcher_round" />
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:paddingTop="@dimen/nav_header_vertical_spacing"
        android:text="@string/nav_header_title"

        android:textAppearance="@style/TextAppearance.AppCompat
        .Body1" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/nav_header_subtitle" />
</LinearLayout>

```

Practical_8_menu_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/action_settings"

```

```

        android:orderInCategory="100"
        android:title="@string/action_settings"
        app:showAsAction="never" />
</menu>

```

JAVA:

Home_fragment & Home_view_model :

```

package com.mk.madpractical.ui.home;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentHomeBinding;
public class HomeFragment extends Fragment {
    private FragmentHomeBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
        savedInstanceState) {
        HomeViewModel homeViewModel =
            new
            ViewModelProvider(this).get(HomeViewModel.class);
        binding = FragmentHomeBinding.inflate(inflater,
            container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textHome;

        homeViewModel.getText().observe(getViewLifecycleOwner(),
            textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class HomeViewModel extends ViewModel {

    private final MutableLiveData<String> mText;

    public HomeViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is home fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Gallery_fragment & Gallery_view_model :


```

package com.mk.madpractical.ui.gallery;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentGalleryBinding;
public class GalleryFragment extends Fragment {
    private FragmentGalleryBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        GalleryViewModel galleryViewModel =
            new
ViewModelProvider(this).get(GalleryViewModel.class);
        binding = FragmentGalleryBinding.inflate(inflater,
            container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textGallery;

galleryViewModel.getText().observe(getViewLifecycleOwner(),

textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class GalleryViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public GalleryViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is gallery fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Slideshow_fragment & / Slideshow_view_model :

```

package com.mk.madpractical.ui.slideshow;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

```

```

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;
import
com.mk.madpractical.databinding.FragmentSlideshowBinding
;
public class SlideshowFragment extends Fragment {
    private FragmentSlideshowBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        SlideshowViewModel slideshowViewModel =
            new
ViewModelProvider(this).get(SlideshowViewModel.class);
        binding = FragmentSlideshowBinding.inflate(inflater,
            container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textSlideshow;

slideshowViewModel.getText().observe(getViewLifecycleOwner(),

textView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
class SlideshowViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public SlideshowViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is slideshow fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}

```

Practical_8.java :

```

package com.mk.madpractical;
import android.os.Bundle;
import android.view.View;
import android.view.Menu;
import com.google.android.material.snackbar.Snackbar;
import
com.google.android.material.navigation.NavigationView;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;

```

```

import androidx.drawerlayout.widget.DrawerLayout;
import androidx.appcompat.app.AppCompatActivity;
import
com.mk.madpractical.databinding.ActivityPractical8Binding;
public class Practical8 extends AppCompatActivity {
    private AppBarConfiguration mAppBarConfiguration;
    private ActivityPractical8Binding binding;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        binding =
        ActivityPractical8Binding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        setSupportActionBar(binding.appBarPractical8.toolbar);
        binding.appBarPractical8.fab.setOnClickListener(new
        View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Snackbar.make(view, "Message function will
                comming soon", Snackbar.LENGTH_LONG)
                    .setAction("Action", null)
                    .setAnchorView(R.id.fab).show();
            }
        });
        DrawerLayout drawer = binding.drawerLayout;
        NavigationView navigationView =
binding.Practical8NavigationView;
        // Passing each menu ID as a set of Ids because each
        // menu should be considered as top level destinations.

```

```

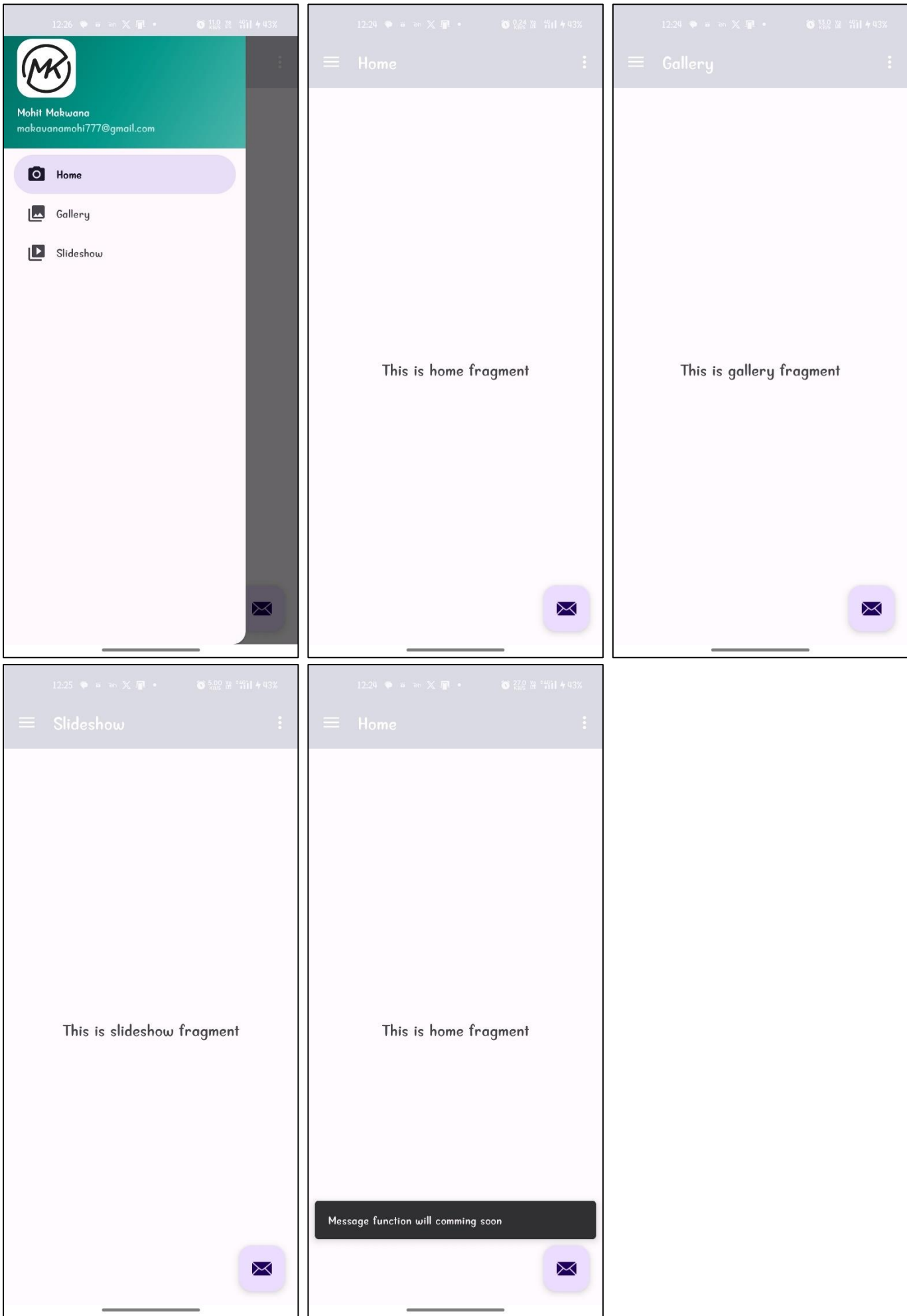
mAppBarConfiguration = new
AppBarConfiguration.Builder(
    R.id.nav_home, R.id.nav_gallery, R.id.nav_slideshow)
    .setOpenableLayout(drawer)
    .build();
    NavController navController =
    Navigation.findNavController(this,
    R.id.nav_host_fragment_content_practical8);
    NavigationUI.setupActionBarWithNavController(this,
    navController, mAppBarConfiguration);
    NavigationUI.setupWithNavController(navigationView,
    navController);
}

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is
        present.
        getMenuInflater().inflate(R.menu.practical8, menu);
        return true;
    }

    @Override
    public boolean onSupportNavigateUp() {
        NavController navController =
        Navigation.findNavController(this,
        R.id.nav_host_fragment_content_practical8);
        return NavigationUI.navigateUp(navController,
mAppBarConfiguration)
        || super.onSupportNavigateUp();
    }
}

```

OUTPUT:



Practical-9

AIM: Develop an Android application that uses the bottom navigation bar to switch between different tabs.

XML:

Practical_layout file:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingTop="?attr/actionBarSize">
    <com.google.android.material.bottomnavigation.
```

BottomNavigationView

```
    android:id="@+id/nav_view"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="0dp"
    android:layout_marginEnd="0dp"
    android:background="?android:attr/windowBackground"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:menu="@menu/bottom_nav_menu" />
</fragment
    android:id="@+id/nav_host_fragment_activity_practical9"
```

```
android:name="androidx.navigation.fragment.NavHostFragment"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:defaultNavHost="true"
    app:layout_constraintBottom_toTopOf="@id/nav_view"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:navGraph="@navigation/mobile_navigation3" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Home_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.home.HomeFragment">
    <TextView
        android:id="@+id/text_home"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Dashboard_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.dashboard.DashboardFragment">
    <TextView
        android:id="@+id/text_dashboard"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Notifications_fragment:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.notifications.NotificationsFragment">
    <TextView
        android:id="@+id/text_notifications"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="8dp"
```

```

        android:layout_marginEnd="8dp"
        android:textAlignment="center"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Bottom_nav_menu:

```

<?xml version="1.0" encoding="utf-8"?>
<menu
xmlns:android="http://schemas.android.com/apk/res/android"
>
    <item
        android:id="@+id/navigation_home"
        android:icon="@drawable/ic_home_black_24dp"
        android:title="@string/title_home" />
    <item
        android:id="@+id/navigation_dashboard"
        android:icon="@drawable/ic_dashboard_black_24dp"
        android:title="@string/title_dashboard" />
    <item
        android:id="@+id/navigation_notifications"
        android:icon="@drawable/ic_notifications_black_24dp"
        android:title="@string/title_notifications" />
</menu>

```

Mobile_navigation:

```

<?xml version="1.0" encoding="utf-8"?>
<navigation
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/mobile_navigation3"
app:startDestination="@+id/navigation_home">
    <fragment
        android:id="@+id/navigation_home"
        android:name="com.mk.madpractical.ui.home.HomeFragment"
        android:label="@string/title_home"
        tools:layout="@layout/fragment_home" />
    <fragment
        android:id="@+id/navigation_dashboard"
        android:name="com.mk.madpractical.ui.dashboard.DashboardFragment"
        android:label="@string/title_dashboard"
        tools:layout="@layout/fragment_dashboard" />
    <fragment
        android:id="@+id/navigation_notifications"
        android:name="com.mk.madpractical.ui.notifications.NotificationsFragment"
        android:label="@string/title_notifications"
        tools:layout="@layout/fragment_notifications" />
</navigation>

```

JAVA:

Practical_9.java :

```

package com.mk.madpractical;
import android.os.Bundle;
import
com.google.android.material.bottomnavigation.BottomNavigationView;
import androidx.appcompat.app.AppCompatActivity;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;
import
com.mk.madpractical.databinding.ActivityPractical9Binding;

public class Practical9 extends AppCompatActivity {
    private ActivityPractical9Binding binding;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        binding =
ActivityPractical9Binding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());
        BottomNavigationView navView =
findViewById(R.id.nav_view);
        // Passing each menu ID as a set of IDs because each
        // menu should be considered as top level destinations.
        AppBarConfiguration appBarConfiguration = new
AppBarConfiguration.Builder(
            R.id.navigation_home, R.id.navigation_dashboard,
            R.id.navigation_notifications)
            .build();
        NavController navController =
Navigation.findNavController(this,
            R.id.nav_host_fragment_activity_practical9);
        NavigationUI.setupActionBarWithNavController(this,
            navController, appBarConfiguration);
        NavigationUI.setupWithNavController(binding.navView,
            navController);
    }
}

```

Home_fragment & Home_view_model :

```

package com.mk.madpractical.ui.home;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.LiveData;
import androidx.lifecycle.MutableLiveData;
import androidx.lifecycle.ViewModel;
import androidx.lifecycle.ViewModelProvider;

```

```

import
com.mk.madpractical.databinding.FragmentHomeBinding;
public class HomeFragment extends Fragment {
    private FragmentHomeBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        HomeViewModel homeViewModel =
            new
ViewModelProvider(this).get(HomeViewModel.class);
        binding = FragmentHomeBinding.inflate(inflater,
container, false);
        View root = binding.getRoot();
        final TextView textView = binding.textHome;
        homeViewModel.getText().observe(getViewLifecycleOwner(), te
xtView::setText);
        return root;
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
}class HomeViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public HomeViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is home fragment");
    }
    public LiveData<String> getText() {
        return mText;
    }
}
}

```

Dashboard_fragment & Dashboard_view_model :

```

package com.mk.madpractical.ui.dashboard;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.ViewModelProvider;
import com.mk.madpractical.databinding.

```

```

FragmentDashboardBinding;

```

```

public class DashboardFragment extends Fragment {
    private FragmentDashboardBinding binding;
    public View onCreateView(@NonNull LayoutInflater inflater,
        ViewGroup container, Bundle
savedInstanceState) {
        DashboardViewModel dashboardViewModel =
            new
ViewModelProvider(this).get(DashboardViewModel.class);

```

```

        binding = FragmentDashboardBinding.inflate(inflater,
container,

```

```

false);

```

```

        View root = binding.getRoot();

```

```

        final TextView textView = binding.textDashboard;

```

```

        dashboardViewModel.getText().observe(getViewLifecycleOwn
er(), textView::setText);

```

```

        return root;

```

```

    } @Override

```

```

    public void onDestroyView() {

```

```

        super.onDestroyView();

```

```

        binding = null;

```

```

    }

```

```

}class DashboardViewModel extends ViewModel {

```

```

    private final MutableLiveData<String> mText;

```

```

    public DashboardViewModel() {

```

```

        mText = new MutableLiveData<>();

```

```

        mText.setValue("This is dashboard fragment");

```

```

    }

```

```

    public LiveData<String> getText() {

```

```

        return mText;

```

```

    }

```

```

}

```

Notifications_fragment & Notifications_view_model :

```

package com.mk.madpractical.ui.notifications;`

```

```

import android.os.Bundle;

```

```

import android.view.LayoutInflater;

```

```

import android.view.View;

```

```

import android.view.ViewGroup;

```

```

import android.widget.TextView;

```

```

import androidx.annotation.NonNull;

```

```

import androidx.fragment.app.Fragment;

```

```

import androidx.lifecycle.ViewModelProvider;

```

```

import

```

```

com.mk.madpractical.databinding.FragmentNotificationsBindi
ng;

```

```

public class NotificationsFragment extends Fragment {

```

```

    private FragmentNotificationsBinding binding;

```

```

    public View onCreateView(@NonNull LayoutInflater inflater,

```

```

        ViewGroup container, Bundle

```

```

savedInstanceState) {

```

```

        NotificationsViewModel notificationsViewModel =

```

```

            new

```

```

ViewModelProvider(this).get(NotificationsViewModel.class);

```

```

        binding = FragmentNotificationsBinding.inflate(inflater,

```

```

container, false);

```

```

        View root = binding.getRoot();

```

```

        final TextView textView = binding.textNotifications;

```

```

        notificationsViewModel.getText().observe(getViewLifecycleOw
ner(), textView::setText);

```

```

        return root;

```

```

    }

```

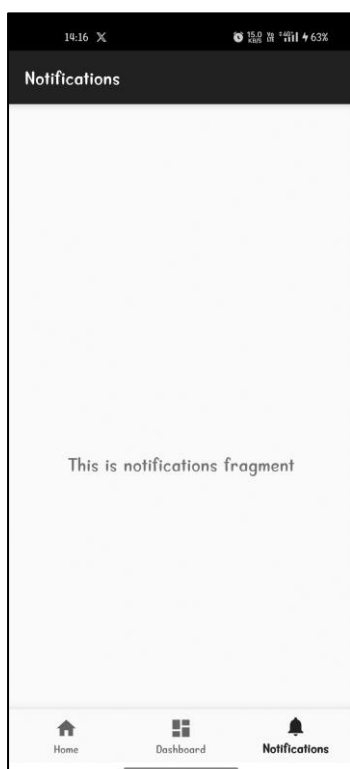
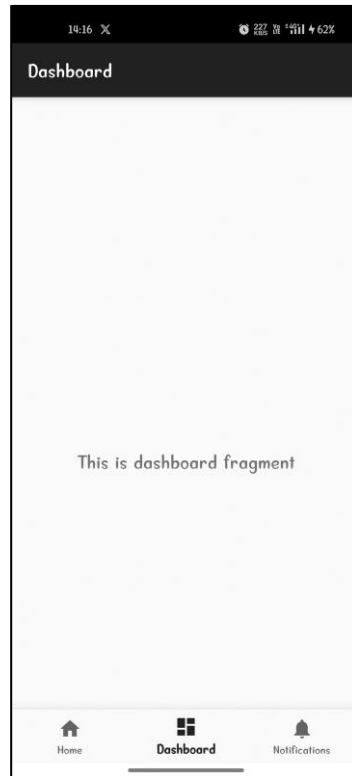
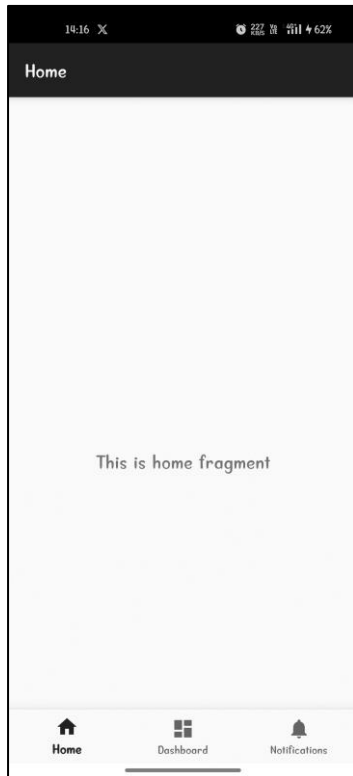
```

@Override
public void onDestroyView() {
    super.onDestroyView();
    binding = null;
}
}class NotificationsViewModel extends ViewModel {
    private final MutableLiveData<String> mText;
    public NotificationsViewModel() {
        mText = new MutableLiveData<>();
        mText.setValue("This is notifications fragment");
    }
}
    }

    public LiveData<String> getText() {
        return mText;
    }
}

```

OUTPUT:



Practical-10

AIM: Develop an Android application that uses an Intent to pass data between different activities

XML:

MainActivity:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

</androidx.constraintlayout.widget.ConstraintLayout>
```

Practical-10:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical_10_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical10">
<TextView
    android:id="@+id/Practical_10_textView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

MainActivity:

```
package com.mk.madpractical;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.Intent;
import android.os.Bundle;
```

```
import android.app.AlertDialog;
import android.content.DialogInterface;
import android.widget.EditText;
import android.widget.LinearLayout;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());

            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        showInputDialog();
    }
    private void showInputDialog() {
        // Create EditText programmatically
        final EditText editTextInput = new EditText(this);
        editTextInput.setHint("Enter your text");
        editTextInput.setLayoutParams(new
        LinearLayout.LayoutParams(
            LinearLayout.LayoutParams.MATCH_PARENT,
            LinearLayout.LayoutParams.MATCH_PARENT));
        new AlertDialog.Builder(this)
            .setTitle("Enter your text")
            .setView(editTextInput)
            .setPositiveButton("OK", new
            DialogInterface.OnClickListener() {
                @Override
                public void onClick(DialogInterface dialog, int
                which) {
                    String inputText =
                    editTextInput.getText().toString();
                    Intent pr10 = new
                    Intent(MainActivity.this, Practical10.class);
                    pr10.putExtra("MSG", inputText);
                    startActivity(pr10);
                }
            })
            .setNegativeButton("Cancel", null)
            .show();
    }
}
```



```

}
@Override
protected void onDestroy() {
    super.onDestroy();
}
}

```

Practical-10:

```

package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.android.material.snackbar.Snackbar;

public class Practical10 extends AppCompatActivity {

```

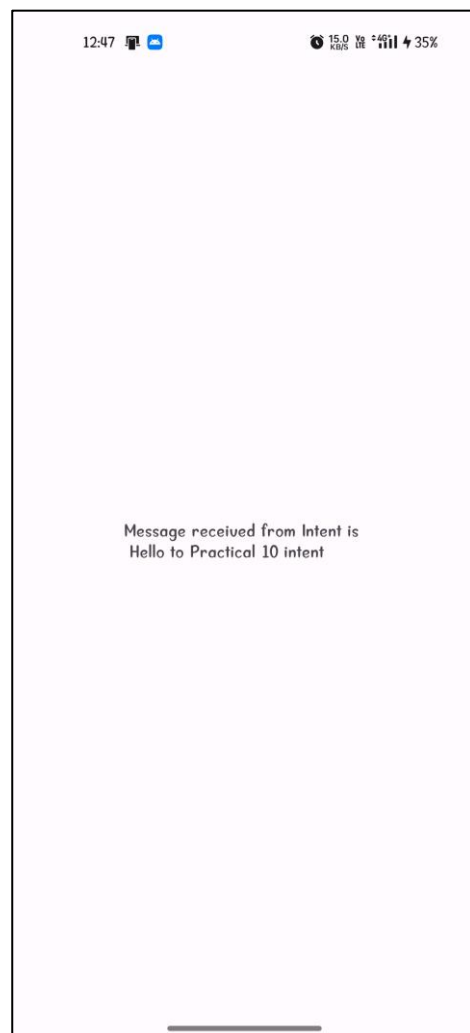
```

    private TextView textView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical10);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical_10_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());

            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        textView =
        findViewById(R.id.Practical_10_textView1);
        Intent intent = getIntent();
        textView.setText("Message received from Intent is
        \n "+intent.getStringExtra("MSG"));
    }
}

```



Practical-11

AIM: Develop an Android application that uses Services to perform background tasks.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical11_layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical11">
<Button
    android:id="@+id/Practical11_start_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start the service"

app:layout_constraintBottom_toTopOf="@id/Practical11_stop_btn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
<Button
    android:id="@+id/Practical11_stop_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Stop the service"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Practical11_start_btn" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Service.java

```
package com.mk.madpractical;

import android.app.Service;
import android.content.Intent;
import android.os.Binder;
import android.os.Handler;
import android.os.Looper;
```

```
import android.os.IBinder;
import android.widget.Toast;

public class Practical11Service extends Service {

    private Thread thread;
    private boolean isServiceRunning;

    public Practical11Service() {
    }

    @Override
    public IBinder onBind(Intent intent) {
        return new Binder() {
            Practical11Service get_service() {
                return Practical11Service.this;
            }
        };
    }

    @Override
    public void onCreate() {
        super.onCreate();
        isServiceRunning = true;

        // Create and start a new thread
        thread = new Thread(new Runnable() {
            @Override
            public void run() {
                while (isServiceRunning) {
                    new
                    Handler(Looper.getMainLooper()).post(new Runnable() {
                        @Override
                        public void run() {
                            Toast.makeText(Practical11Service.this,
                            "Service is running", Toast.LENGTH_SHORT).show();
                        }
                    });
                }
            }
        });
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int
```

```
startId) {
    if (!thread.isAlive()) {
        thread.start(); // Start the thread
    }
    return START_STICKY; // Keep the service running
}
```

```
@Override
public void onDestroy() {
    super.onDestroy();
    isServiceRunning = false; // Stop the loop
    if (thread != null && thread.isAlive()) {
        thread.interrupt(); // Safely stop the thread
    }
}
```

Practical11.java

```
package com.mk.madpractical;
```

```
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class Practical11 extends AppCompatActivity {
    Button start,stop;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
EdgeToEdge.enable(this);
setContentView(R.layout.activity_practical11);
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical11_layout), (v, insets) -> {
    Insets systemBars =
    insets.getInsets(WindowInsetsCompat.Type.systemBars())
});
```

```
v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
return insets;
});
```

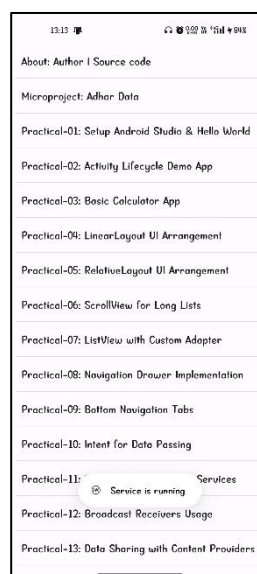
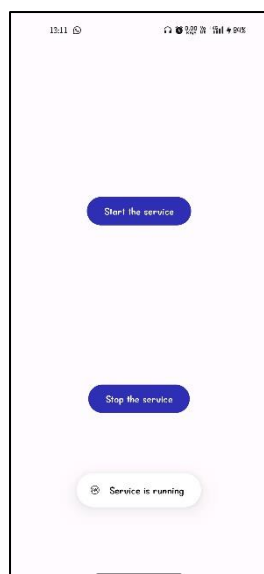
```
start = findViewById(R.id.Practical11_start_btn);
stop = findViewById(R.id.Practical11_stop_btn);
Intent service = new
Intent(this,Practical11Service.class);
```

```
View.OnClickListener listener = new
View.OnClickListener() {
```

```
@Override
public void onClick(View v) {
```

```
try {
    if (v.equals(start)){
        startService(service);
    }else {
        stopService(service);
    }
} catch (Exception e) {
    throw new RuntimeException(e);
}
}
```

```
};
start.setOnClickListener(listener);
stop.setOnClickListener(listener);
}
```



Practical-12

AIM: Develop an Android application that uses Broadcast Receivers to receive and handle system-level broadcasts.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical12_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical12_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="plug the power cable or take it off to
see the broadcast"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.IntentFilter;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
```

```
public class Practical12 extends AppCompatActivity {
    Practical12_broadcast broadcast;
    IntentFilter intentFilter;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical12);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical12_ConstraintLayout), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

        textView = findViewById(R.id.Practical12_textview);

        broadcast = new Practical12_broadcast();
        intentFilter = new IntentFilter();

        intentFilter.addAction("android.intent.action.ACTION_POWER_CONNECTED");

        intentFilter.addAction("android.intent.action.ACTION_POWER_DISCONNECTED");
        registerReceiver(broadcast, intentFilter,
Context.RECEIVER_NOT_EXPORTED);

    }

    @Override
    protected void onDestroy() {
        unregisterReceiver(broadcast);
        super.onDestroy();
    }

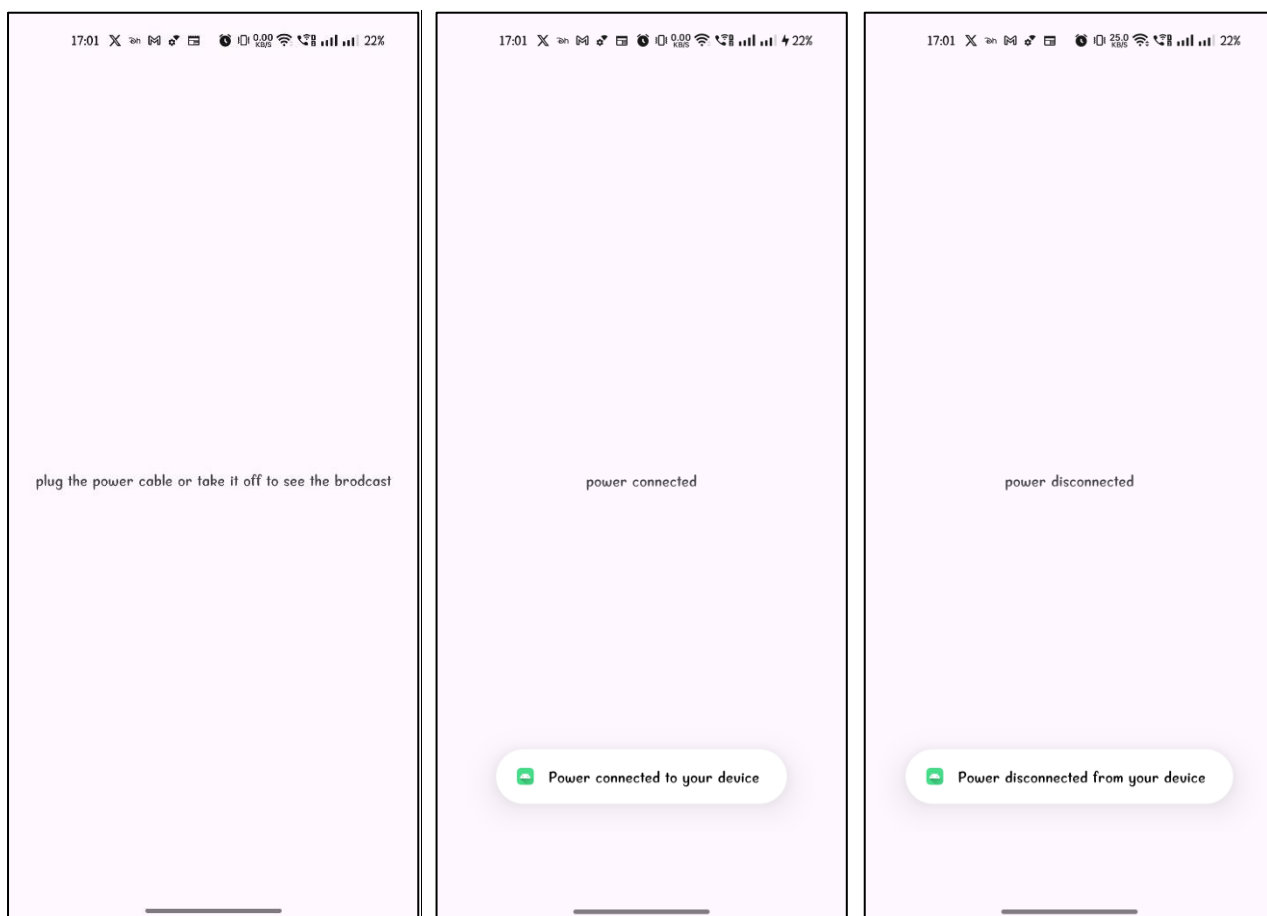
    class Practical12_broadcast extends BroadcastReceiver {
        @Override
        public void onReceive(Context context, Intent intent)
```

```

{
    String action = intent.getAction();

    if (action != null) {
        switch (action) {
            case
"android.intent.action.ACTION_POWER_CONNECTED":
                Toast.makeText(context, "Power connected
to your device", Toast.LENGTH_SHORT).show();
                textView.setText("power connected");
                break;
            case
"android.intent.action.ACTION_POWER_DISCONNECTED
":
                Toast.makeText(context, "Power
disconnected from your device",
Toast.LENGTH_SHORT).show();
                textView.setText("power disconnected");
                break;
            default:
                Toast.makeText(context, "Unknown action:
" + action, Toast.LENGTH_SHORT).show();
                textView.setText("unknown action
triggered");
                break;
        }
    }
}

```



Practical-13

AIM: Develop an Android application that uses Content Providers to share data between different apps and components.

XML:

Manifest file:

Add this to Manifest.xml

```
<provider
    android:name=".Pr13ContentProvider"
    android:authorities="com.mk.madpractical"
    android:enabled="true"
    android:exported="true" />
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical13_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical13_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="we are trying to receive context from the context
provider"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <Button
        android:id="@+id/Practical13_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Practical13_textview"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Pr13ContentProvider.java:

```
package com.mk.madpractical;

import android.content.ContentProvider;
import android.content.ContentValues;
import android.database.Cursor;
```

```
import android.database.MatrixCursor;
import android.net.Uri;

public class Pr13ContentProvider extends ContentProvider {

    public Pr13ContentProvider() {
    }

    @Override
    public int delete(Uri uri, String selection, String[] selectionArgs) {
        // Implement this to handle requests to delete one or more rows.
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public String getType(Uri uri) {
        // at the given URI.
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public Uri insert(Uri uri, ContentValues values) {
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public boolean onCreate() {
        return false;
    }

    @Override
    public Cursor query(Uri uri, String[] projection, String selection,
String[] selectionArgs, String sortOrder) {
        // Create a Cursor with a single column containing the greeting
message
        MatrixCursor cursor = new MatrixCursor(new
String[]{"message"});
        cursor.addRow(new String[]{"Hello from the MAD Practical 12
Content Provider!"});
        return cursor;
    }

    @Override
    public int update(Uri uri, ContentValues values, String selection,
String[] selectionArgs) {
        throw new UnsupportedOperationException("Not yet
implemented");
    }
}
```

Practical13.java:

```
package com.mk.madpractical;
import android.content.ContentResolver;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```

import android.net.Uri;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.activity.EdgeToEdge;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical13 extends AppCompatActivity {
    Button btn;
    TextView textView;
    private static final Uri CONTENT_URI =
Uri.parse("content://com.mk.madpractical/hello");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical13);

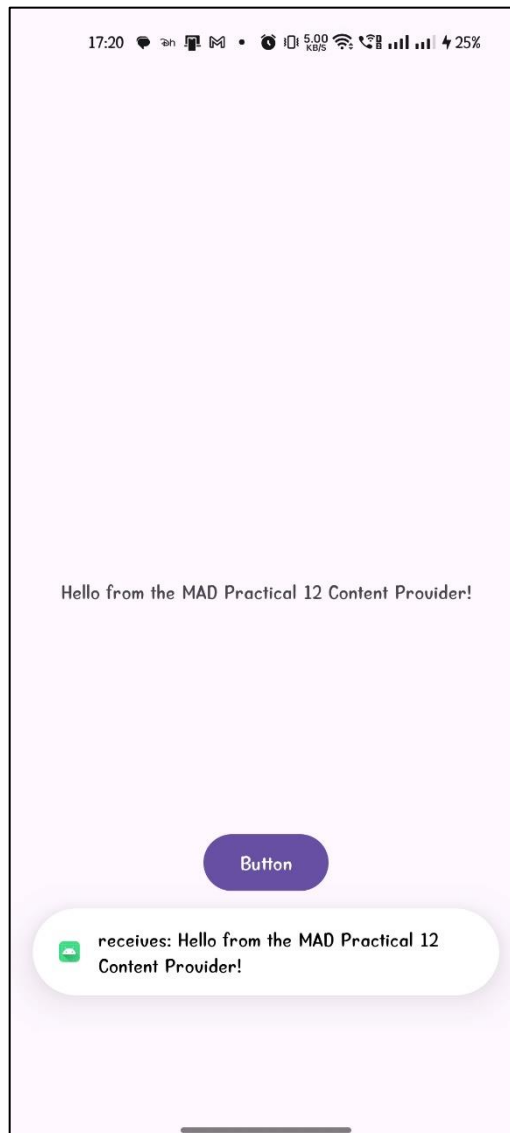
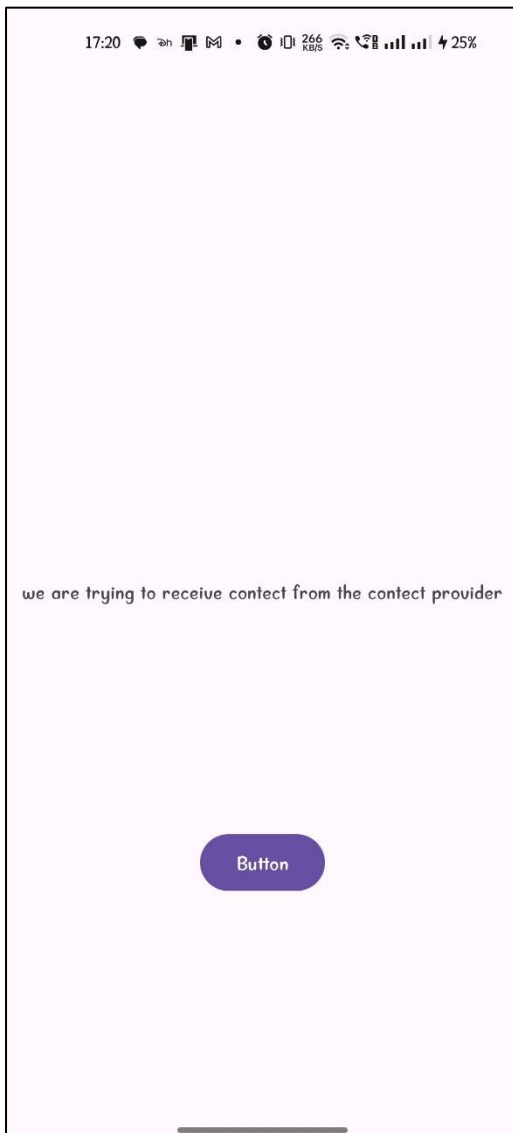
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical13_ConstraintLayout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,

```

```

systemBars.right, systemBars.bottom);
        return insets;
    });
    btn = findViewById(R.id.Practical13_button);
    textView = findViewById(R.id.Practical13_textview);
    btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            ContentResolver contentResolver = getContentResolver();
            Cursor cursor = contentResolver.query(CONTENT_URI, null,
null, null, null);
            if (cursor != null && cursor.moveToFirst()) {
                String msg =
cursor.getString(cursor.getColumnIndex("message"));
                textView.setText(msg);
                Toast.makeText(Practical13.this, "receives: "+msg,
Toast.LENGTH_SHORT).show();
                cursor.close();
            }
        }
    });
}
}

```



AIM: Develop an Android application that uses Content Providers to read system-level data, such as contacts and calendar events.

XML:

Manifest file:

Add this to Manifest.xml

```
<uses-permission  
  android:name="android.permission.READ_CONTACTS" />
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical14_ConstraintLayout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical14">

<ListView
    android:id="@+id/Practical14_listview"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.5"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
o package com.mk.madpractical;
```

```
import android.Manifest;
import android.content.ContentResolver;
import android.content.pm.PackageManager;
import android.database.Cursor;
import android.net.Uri;
import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.ArrayList;

public class Practical14 extends AppCompatActivity {

    private static final int PERMISSION_REQUEST_CODE
```

```

private ListView contactsListView;
private ArrayList<String> contactsList;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_practical14);

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical14_ConstraintLayout), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
        return insets;
    });

    contactsListView = findViewById(R.id.Practical14_listview);
    contactsList = new ArrayList<>();

    if (ContextCompat.checkSelfPermission(this,
Manifest.permission.READ_CONTACTS)
!= PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.READ_CONTACTS},
PERMISSION_REQUEST_CODE);
    } else {
        loadContacts();
    }
}

private void loadContacts() {
    ContentResolver contentResolver = getContentResolver();
    Uri uri =
android.provider.ContactsContract.CommonDataKinds.Phone.CONTENT_URI;

    Cursor cursor = contentResolver.query(uri, null, null, null, null);

    if (cursor != null) {
        while (cursor.moveToNext()) {
            String name =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.DISPLAY_NAME));
            String phoneNumber =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.NUMBER));
            contactsList.add(name + ": " + phoneNumber);
        }
        cursor.close();
    }

    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, contactsList);
    contactsListView.setAdapter(adapter);
}

@Override
public void onRequestPermissionsResult(int requestCode,

```



```

    } else {
        Toast.makeText(this, "Permission denied to read contacts",
Toast.LENGTH_SHORT).show();
    }
}
}
}
}

```



Practical-10

AIM: Develop a simple app that demonstrates the activity lifecycle.

XML:

onStart

JAVA:

onStart